



Government API developer portal, registry, sandbox, and oversight layer

A working prototype for discovering MDA-owned APIs, registering new services, managing OpenAPI documentation and versions, requesting governed access, testing integrations, and auditing inter-agency data exchange.

The system is designed for the Ministry of ICT and National Guidance innovator showcase. It demonstrates governed interoperability without connecting to live NIRA, URA, URSB, NITA-U, MoWT, or other production government systems.

[API catalog and registry](#)[OpenAPI documentation](#)[Version control](#)[Sandbox execution](#)[Access approvals](#)[Audit and analytics](#)



1. What The Platform Provides

GovHub API models a central interoperability control plane around government APIs. It gives MDAs one place to publish services, document contracts, request lawful access, test integrations with mock responses, and prove that data exchange is governed.

5**Seeded government APIs**

Identity verification, tax status, business lookup, driving permit verification, and composite eligibility.

4**Primary platform roles**

Developer, API owner, compliance reviewer, and platform administrator, with account approval states.

1**Governed developer portal**

Catalog, docs, sandbox, registry, access matrix, analytics, and audit views in one prototype.

Audience And Purpose

- MoICT can demonstrate a repeatable national API governance model.
- MDAs can expose APIs with ownership, statutory basis, lifecycle status, and security classification.
- Developers can find APIs, review docs, request sandbox access, and test integrations.
- Reviewers can inspect permissions, audit logs, compliance status, and data-sharing activity.

Implemented Technical Shape

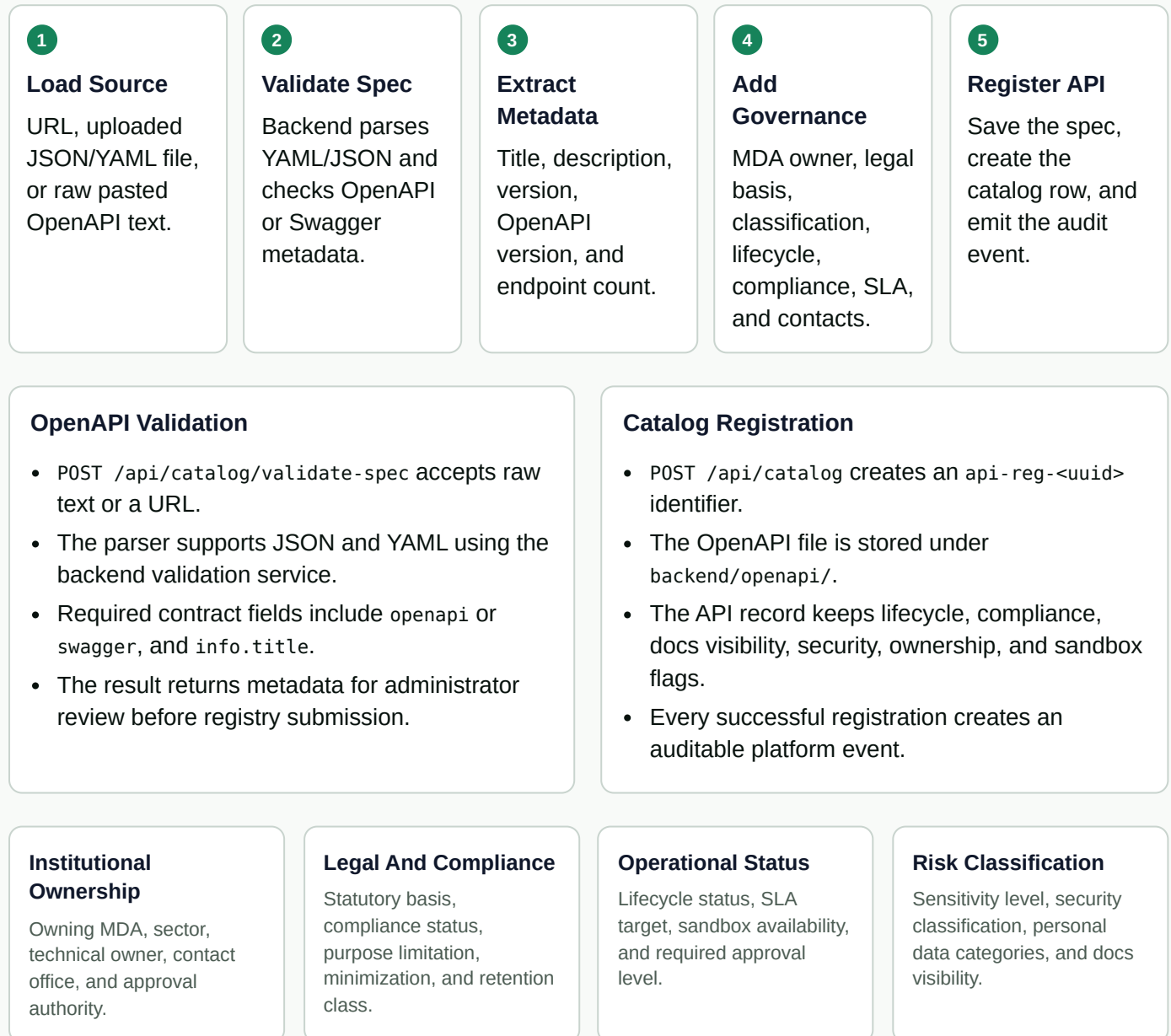
- Frontend: React, TypeScript, Vite, React Router, and GovHub UI components.
- Backend: Node.js, Express, TypeScript, and PostgreSQL through pg.
- Contracts: OpenAPI files stored under backend/openapi/.
- Security: sessions, RBAC, MFA support, encrypted sensitive fields, TLS/HSTS support, and audit logs.

LAYER	RESPONSIBILITY	VISIBLE IN DEMO
Catalog	Searchable MDA API inventory with governance metadata.	/ and API detail pages
Registry	Validates and registers new OpenAPI-backed services.	/catalog/add
Docs	Renders authorized API documentation and downloadable contracts.	/docs and /docs/:apiId
Operations	Approvals, keys, access matrix, audit logs, and analytics.	/dashboard



2. Creating, Validating, And Publishing APIs

The Add API workflow lets an authorized administrator or API owner onboard a new ministry API from a formal OpenAPI source, then attach the governance data needed for public-sector interoperability.



Registry Outcome

New APIs become governed catalog assets with a machine-readable contract, accountable owner, explicit data-use purpose, lifecycle state, compliance position, and audit trail.



3. Managing API Documentation And Versions

Documentation is generated from formal OpenAPI contracts and exposed through dedicated GovHub routes. Visibility is controlled per API, so technical reference material can be public, authenticated, or restricted to approved access groups.

Documentation Experience

- `/docs` lists only APIs visible to the current visitor.
- `/docs/:apiId` renders a single API's OpenAPI documentation.
- API detail pages link to the same technical reference while keeping governance and sandbox workflow visible.
- Docs include endpoint groups, schemas, examples, downloads, and generated cURL, JavaScript, Python, and Java samples.

Documentation Visibility

- **public** visible without sign-in.
- **authenticated** visible to approved signed-in users.
- **restricted** visible to admins, reviewers, owning API owners, and approved consumers.
- Admins and owning API owners can update visibility when they can manage the API.

VERSION CAPABILITY	HOW IT WORKS	GOVERNANCE VALUE
Version metadata	Each spec records semantic version, OpenAPI version, endpoint count, SHA hash, status, notes, and creation time.	Reviewers can see what changed and which contract is active.
Current version selection	The catalog points to the current OpenAPI spec while previous records remain available as non-current versions.	Consumers integrate against a known contract instead of informal endpoint behavior.
Spec reconciliation	Existing catalog specs are backfilled into <code>api_versions</code> when version records are missing.	Legacy and newly registered APIs share the same version model.
Deletion control	The demo supports deleting non-current versions while protecting the current version from accidental removal.	Operational cleanup does not break active documentation or sandbox contracts.

Downloadable Assets

Authorized users can download OpenAPI JSON or YAML assets. Restricted assets are disclosed only after the docs access decision succeeds.

Machine-Readable Contracts

The platform demonstrates that ministry API documentation can be treated as a governed contract, not a manually maintained webpage.



4. Security, Sandbox Governance, And Demo Readiness

GovHub API separates visibility from operational access. A user may browse public metadata, but sandbox execution, approvals, keys, audit logs, and administrative workflows require an approved account and the correct role.

Account Control

- Public, pending, approved, rejected, and suspended states.
- Roles: developer, API owner, reviewer, admin.
- MFA support and account settings.

Access Workflow

- Purpose, legal basis, requested fields, environment, and volume tier.
- API-owner or admin approval.
- Scoped sandbox key with expiry, revocation, and deletion.

Sandbox Enforcement

- Deterministic mock services.
- Structured errors and correlation IDs.
- Denied calls for missing, expired, revoked, or wrong-scope keys.

Dashboard

Approvals, accounts, access matrix, audit trails, analytics, and oversight views.

Audit Trail

Access requests, approvals, key lifecycle events, API registration, and sandbox activity.

Access Matrix

Active approved data-sharing channels between consumers and source APIs.

Security Controls

RBAC, sessions, encryption at rest, TLS/HSTS support, rate limiting, and privacy metadata.

Seeded Demo APIs

NIRA Identity Verification API

URA Tax Compliance Status API

URSB Business Registration Lookup

MoWT Driving Permit Verification API

Service Uganda Composite Eligibility

Presenter Flow

- Discover an API, inspect governance, and request sandbox access.
- Approve the request as admin or API owner and issue a scoped key.
- Run an allowed sandbox request, then show a denied request.
- Open the access matrix, audit logs, and analytics for oversight.
- Register a draft API from an OpenAPI specification.

Current Readiness

Ready for local demo: seeded accounts, catalog, docs, OpenAPI validation, registration, versions, sandbox, approvals, keys, audit, analytics, e2e, and accessibility checks.

Production Boundary

Uses mock integrations only. Production adoption requires live agency integration, hardening, penetration testing, SLAs, agreements, and deployment governance.